When Should LLM Judges Abstain? Multi-View Protocol Instability as a Per-Instance Signal for Selective LLM Judging

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Abstract

Large language models are increasingly used as automated judges to evaluate generated text, yet these judges are prone to overconfidence, positional bias, and sensitivity to evaluation-protocol wording. We propose **CARE-Judge** (Calibrated, Abstaining, and Robust Evaluation), a selective evaluation framework that determines *when* an LLM judge's verdict should be trusted. Our insight is *protocol-stability* used as an asymmetric signal: when a judge's verdict changes under semantically equivalent evaluation protocols—rubric paraphrases, response-order permutations, and stochastic resampling—the verdict is often unreliable, whereas stability alone is *not* evidence of correctness, since a judge can be stably wrong. CARE-Judge fuses these protocol-stability features into a learned correctness calibrator and pairs it with an inherited empirical selective-risk procedure (a held-out calibration threshold certified by an exact Clopper–Pearson bound, validated by measured risk-violation rates rather than a new theorem). Across three primary judge models (Qwen2.5-1.5B, DeepSeek-V4, GPT-5.5) and four benchmarks (JudgeBench, TL;DR, RewardBench, LMArena) totaling over 22,000 pairwise evaluations, protocol-stability signals separate correct from incorrect verdicts by up to 27 accuracy points for mid-capability judges. At a matched six-call budget the fused calibrator attains the best or statistically-tied AUROC on seven of eight DeepSeek/GPT-5.5 benchmark pairs (tying the already-saturated base confidence on the eighth), beating self-consistency, rubric, the SCOPE-style bidirectional signal, and a Trust-or-Escalate agreement baseline; on GPT-5.5 it reaches AUROC 0.85–0.99, and on the mid-capability DeepSeek-V4 fusion adds up to $+6.3$ AUROC points over the best single signal. It accepts 82% of GPT-5.5's RewardBench verdicts at 88% accuracy (raw 71%) and 100% of DeepSeek's at 93%. Critically, when the judge is near-random (Qwen-1.5B, 49% on JudgeBench) 65% of its errors are stable-but-wrong and CARE-Judge abstains entirely—a boundary condition we treat as a finding rather than a success.

1 Introduction

Large language models (LLMs) are now widely deployed as automated judges—scoring RLHF outputs (Ouyang et al. 2022), ranking chatbot responses (Zheng et al. 2023), and evaluating generation quality—but they make systematic errors, exhibit positional biases, and vary verdicts based on evaluation-protocol wording, all while expressing high confidence (Li et al. 2024; Wang et al. 2023; Wei et al. 2024).

Prior work has established that LLM judges are overconfident and has begun to control this risk explicitly. Jung, Brahman, and Choi (2025) showed that standard confidence measures (predictive probability, verbalized confidence) poorly predict whether a judge's verdict agrees with human annotators, and proposed *Simulated Annotators* with cascaded selective evaluation to provide provable human-agreement guarantees; Badshah, Emami, and Sajjad (2026) instead aggregate both response orderings into a permutation-invariant conformal score. These methods each rely on a *single* confidence source—simulated-annotator agreement or bidirectional preference probability—and do not exploit the broader observation that sensitivity across *multiple* semantically equivalent evaluation protocols is itself an error signal. We build directly on their selective-risk machinery (a held-out calibration threshold with an exact Clopper–Pearson bound) and do not claim it as a contribution; our contribution is the confidence signal that machinery is applied to. We are careful *not* to over-claim its formal guarantees: because our acceptance threshold is chosen data-dependently on the calibration split, we treat the risk criterion as an empirical target validated by measured violation rates, not as a family-wise-corrected sequential test.

We investigate a complementary and deliberately asymmetric hypothesis: **a judgment that is unstable under semantically equivalent evaluation protocols is often unreliable, whereas stability alone is not sufficient evidence of correctness.** Concretely, if an LLM judge produces different rankings when (i) the rubric is paraphrased, (ii) the response order is swapped, or (iii) the judgment is re-sampled at nonzero temperature, the resulting verdict is less trustworthy. The converse does not hold: a judge can be stably wrong, applying the same erroneous heuristic (e.g., a length or fluency preference) under every protocol variant (Xu et al. 2026). We call this property *protocol stability*, and we show that protocol *instability* is a useful sufficient warning signal in competence regimes where the judge is at least moderately accurate—and we characterize precisely where it fails.

We organize our study around three questions: *(RQ1)* does per-instance protocol stability predict judge error?

(RQ2) does fusing protocol-stability signals improve the risk–coverage tradeoff over standard confidence and single-signal baselines at a matched budget? *(RQ3)* in which capability and task regimes do these signals help, and where do they break down?

Building on this insight, we introduce **CARE-Judge** (Calibrated, Abstaining, and Robust Evaluation), which (i) collects multi-signal protocol-stability features—rubric perturbation, position-swap consistency, and stochastic self-consistency—for each pairwise judgment; (ii) learns a correctness calibrator $\hat{p}(\text{correct} \mid \text{features})$ on a training split; (iii) selects an acceptance threshold on a disjoint calibration split by an empirical prefix search certified with an exact Clopper–Pearson bound (Jung, Brahman, and Choi 2025; Clopper and Pearson 1934); and (iv) reports selective risk, coverage, and empirical risk-violation rates on a held-out test split.

We validate CARE-Judge across three judge models spanning a wide capability range (Qwen2.5-1.5B, DeepSeek-V4, GPT-5.5) and four benchmarks covering objective correctness (JudgeBench (Tan et al. 2024)), reward-model preference (RewardBench (Lambert et al. 2024)), summarization preference (TL;DR (Stiennon et al. 2020)), and open-ended human preference (LMArena (Zheng et al. 2023)), totaling over 22,000 pairwise evaluations. Our experiments answer the three research questions directly: protocol instability produces accuracy gaps of up to 27 points between stable and unstable subsets where the judge is at least moderately accurate but loses its predictive sign near chance (RQ1); at a matched six-call budget the fused calibrator attains the best or statistically-tied AUROC in 9 of 12 judge–dataset pairs, adding up to $+6.3$ AUROC points over the best single signal on the mid-capability DeepSeek-V4 (RQ2); and below a competence floor the signal *inverts*, so that Qwen2.5-1.5B (49.4% on JudgeBench) produces stable-but-wrong verdicts on which CARE-Judge correctly abstains (RQ3).

Our contributions are:

1. **Competence-gated asymmetry (primary).** We establish—and quantify—that protocol instability predicts judge error only above a minimum judge-competence threshold, and that below it the signal *inverts*: a near-random judge produces stable-but-wrong verdicts (Section 4.4), a boundary we validate with a negative-control experiment showing that on a mid-capability judge non-equivalent rubrics flip $2.2\times$ more often than equivalent ones, whereas a near-random judge is insensitive to the distinction. To our knowledge this competence-gated asymmetry has not been characterized in prior per-instance judge-reliability work.
2. **Decision-oriented multi-view fusion.** We fuse three families of protocol-perturbation instability—rubric paraphrase, response permutation, and stochastic resampling—into a *single per-instance accept/abstain decision under an explicit risk budget*, and show it ranks correct versus incorrect verdicts better than standard confidence, single-signal baselines (including the SCOPE-style bidirectional signal), and ensemble aggregation at a matched inference budget (Section 4). Prompt- and order-sensitivity have been studied before as *diagnostic* properties (Hua et al. 2025; Guan et al. 2025; Anghel et al. 2025a; Huang, Sun, and Yang 2026), and individual consistency signals have been used to *rank* or ensemble judges (Li et al. 2025b; Gupta and Kumar 2026); our contribution is neither observing protocol sensitivity nor a single consistency check, but combining all three perturbation families into one calibrated, risk-controlled acceptance rule (Section 3).

We do not claim the risk-control machinery as novel; it is adopted from prior selective-judging work (Jung, Brahman, and Choi 2025; Badshah, Emami, and Sajjad 2026). Code and data will be released upon acceptance.

2 Related Work

LLM-as-a-Judge. The use of LLMs as automated evaluators has grown rapidly. Zheng et al. (2023) introduced Chatbot Arena and MT-Bench, establishing pairwise LLM judging as a scalable evaluation paradigm. Li et al. (2024) provided a comprehensive survey of LLM-based evaluation methods, identifying biases including position bias, length bias, and self-enhancement bias. Wang et al. (2023) proposed mitigating position bias via multi-turn debate and swap-and-compare. Wei et al. (2024) systematically evaluated LLM-as-a-judge prompt sensitivity, finding that rubric template diversity affects alignment scores. Recent work also cautions that surface agreement need not reflect substantive quality: Xu et al. (2026) give a mechanistic-interpretability account showing that judge biases arise from shared internal heuristics, which directly motivates our position that stability is a warning signal but not a correctness guarantee, and that a judge's self-reported confidence can be unfaithful to its underlying decision—motivating our use of perturbation-based stability rather than verbalized confidence alone. Li et al. (2025a) further document preference leakage as a contamination mode in LLM-as-a-judge pipelines, underscoring the need for strictly disjoint data splits when calibrating any judge-reliability estimator.

Prompt-Sensitivity, Consistency, and Per-Instance Judge Reliability. A growing body of work studies how LLM outputs shift under semantically equivalent prompt, rubric, or ordering perturbations, and how such shifts bear on *per-instance* reliability. Hua et al. (2025) ask whether paraphrase sensitivity is a flaw or an artifact; Guan et al. (2025) isolate the input-order effect; Anghel et al. (2025a) build a pairwise meta-evaluator diagnosing judge instability; and Huang, Sun, and Yang (2026) use prompt perturbation to stabilize pairwise evaluation. Closer to the per-instance regime, Gupta and Kumar (2026) diagnose judge reliability with conformal prediction sets, Choi et al. (2026) apply item-response theory, and Li et al. (2025b) ensemble prompt variants to raise accuracy. Ling et al. (2025) caution that abstention can itself be a prompt artifact—a caveat we address via competence-gating (Section 4.4).

We do *not* claim to be the first to observe protocol sensitivity or to use a consistency signal; we differ in *use* and *scope*. In use, prior work diagnoses judges globally (Hua et al. 2025; Guan et al. 2025; Anghel et al. 2025a), builds per-instance reliability *sets* for analysis (Gupta and Kumar 2026; Choi et al. 2026), or ensembles prompts to *improve accuracy* (Li et al. 2025b; Huang, Sun, and Yang 2026); we instead fuse three perturbation families into a single calibrated accept/abstain decision under an explicit risk budget. In scope, none characterizes the competence-gated asymmetry we report—instability predicts error above a competence threshold and inverts below it.

Selective Prediction and Abstention. Selective classification (Geifman and El-Yaniv 2017; El-Yaniv and Wiener 2010) enables a predictor to abstain on uncertain inputs; Wen et al. (2025) survey abstention in LLMs (verbalized confidence, self-consistency, retrieval), Krishnan, Khanna, and Tickoo (2024) propose uncertainty-calibrated fine-tuning using hidden states, and Tayebati et al. (2025) learn conformal abstention policies for adaptive risk management. Our work differs by focusing on *evaluation-protocol stability* as the uncertainty source, which is specific to the judging task and does not require model internals or a learned abstention controller.

Risk-Controlled Selective Judging. Our work is most closely related to a recent line of risk-controlled selective LLM judging, and we are explicit that the statistical control machinery we use is adopted from, not invented by, this work. Jung, Brahman, and Choi (2025) (Trust-or-Escalate) proposed cascaded selective evaluation for LLM judges with human-agreement guarantees, using Simulated Annotators as the confidence source together with confidence-threshold calibration and Clopper–Pearson risk control. Concurrently, Badshah, Emami, and Sajjad (2026) (SCOPE) proposed selective conformal pairwise judging, aggregating the probabilities of both response orderings into a permutation-invariant score and deriving risk–coverage guarantees on benchmarks including MT-Bench. Sheng et al. (2025) likewise applied conformal prediction to quantify LLM-as-a-judge uncertainty through interval evaluations. CARE-Judge uses a comparable selective-risk apparatus—selecting an acceptance threshold on a held-out calibration split via an exact Clopper–Pearson upper bound—rather than proposing new statistical guarantees; we treat this apparatus as an inherited component and are careful (Section 3) to describe exactly the empirical, calibration-conditional form of risk control we use. Our contribution is orthogonal and lies in the *confidence signal*: whereas Trust-or-Escalate relies on simulated-annotator agreement and SCOPE on bidirectional preference probability alone, we combine rubric-paraphrase instability, position-swap inconsistency, and stochastic self-consistency into a single learned calibrator, and we characterize the judge-capability regimes in which such protocol-stability signals help or fail. Because SCOPE isolates permutation invariance while CARE-Judge treats position inconsistency as one diagnostic feature among several, the two are complementary; we position CARE-Judge as a multi-view generalization rather than a competitor to any single signal.

Judge Benchmarks and Rubric-Based Evaluation. We use JudgeBench (Tan et al. 2024) and RewardBench (Lambert et al. 2024) as hard, objectively-labeled benchmarks and TL;DR (Stiennon et al. 2020) and LMArena (Zheng et al. 2023) as subjective-preference benchmarks. Prior rubric work treats rubrics as fixed criteria to align with—Hong et al. (2026) align judges with human scoring rubrics and Anghel et al. (2025b) propose a rubric-driven multi-metric framework—whereas we treat *rubric sensitivity* (how much a verdict changes under semantically equivalent paraphrases) as an uncertainty signal. We are not aware of prior work that uses rubric-paraphrase instability specifically as a per-instance acceptance signal within a risk-controlled selective-judging framework; we frame this narrower claim as our contribution and validate it empirically.

Cost-Aware Cascades. Routing easy queries to cheap models and escalating hard ones is a well-established cost-reduction strategy; Zellinger, Liu, and Thomson (2025) study LLM cascades with early abstention and formal cost analysis. Our exploratory cascade experiment (Appendix) applies the same idea to judge routing and defers a formal cascade-level guarantee to future work.

3 Method

CARE-Judge converts an LLM judge f into a *selective evaluator* $(\hat{f}, \hat{c})$ that either outputs a judgment $\hat{f}(x)$ or abstains, based on a confidence measure $\hat{c}(x)$ and a risk-controlled threshold $\hat{\lambda}$. The framework has three components: (1) multi-signal protocol-stability feature extraction, (2) learned correctness calibration, and (3) empirical risk-controlled threshold selection with an exact Clopper–Pearson bound, validated by measured risk-violation rates rather than a new finite-sample theorem.

3.1 Protocol-Stability Features

Given a pairwise evaluation item $x = (q, a_1, a_2)$ consisting of a prompt q and two candidate responses, we issue multiple judge calls under semantically equivalent evaluation protocols and measure the stability of the resulting verdicts. Let $f(x; r, \tau)$ denote the judge's output (winner $\in \{A, B\}$ and confidence $\in [0, 1]$) given rubric r and sampling temperature τ.

Rubric Perturbation Stability. We define a set of $K = 3$ semantically equivalent rubric variants $\{r_1, \ldots, r_K\}$ that are intended to express the same evaluation criteria in different wording (e.g., "Prefer the more correct response" vs. "Choose the answer a careful evaluator would prefer"); all variants are listed verbatim in the Appendix. For each item, we collect verdicts $\{v_1, \ldots, v_K\}$ where $v_k = f(x; r_k, 0)$. We compute:

$$\text{rubric\_vote\_share} = \max_{w \in \{A,B\}} \frac{|\{k : v_k = w\}|}{K}, \quad (1)$$

$$\text{rubric\_flip} = \mathbb{1}[|\{v_1, \ldots, v_K\}| > 1], \quad (2)$$

where rubric_flip $= 1$ if any rubric variant produces a different winner. The intuition is that a verdict that changes under paraphrase is *protocol-dependent* and not robust to the stated specification. We avoid the stronger claim that it is "content-irrelevant," since a paraphrase can also shift the weight a judge places on competing criteria; a negative-control experiment with intentionally non-equivalent rubrics

(Appendix B) confirms that on a mid-capability judge, non-equivalent rubrics produce $2.2\times$ higher flip rates than equivalent ones, validating that equivalent-rubric flips reflect genuine instability—while on a near-random judge the two conditions are indistinguishable, reinforcing that the signal should not be trusted below the competence threshold.

Position-Swap Consistency. We judge the item in both response orderings: (q, a_1, a_2) and (q, a_2, a_1). Let $v_{\text{orig}} = f(x; r_1, 0)$ and $v_{\text{swap}} = f(\text{swap}(x); r_1, 0)$, where swap reverses the response order. We map v_{swap} back to the original coordinate system and compute:

$$\text{swap\_consistency} = \mathbb{1}[\text{map}(v_{\text{swap}}) = v_{\text{orig}}]\,. \tag{3}$$

This probes positional bias: a judge that reverses its verdict under response reordering is unreliable.

Self-Consistency. We form a set of S judgments under the base rubric that pools the deterministic base verdict (τ=0) with $S-1$ stochastic re-samples at temperature $\tau > 0$, and compute the majority vote share:

$$\text{self\_vote\_share} = \max_{w} \frac{|\{s : f(x; r_1, \tau_s) = w\}|}{S}. \tag{4}$$

In all reported experiments $S = 3$ (one τ=0 base verdict and two τ=0.7 samples), so the score is genuinely informative rather than degenerate. An optional *adaptive-k* early-stopping rule (self_vote_share $\geq \tau_{\text{stop}}$ after a minimum of two samples) is available to reduce cost but was disabled in the main experiments.

Feature Vector. CARE-Judge uses six raw judge calls per item—one base (τ=0), two self-consistency samples (τ=0.7), one position swap, and two additional rubric paraphrases (the base rubric doubling as the third rubric verdict)—and forms a 13-dimensional feature vector grouped into three families (full definitions in Appendix D, Table 6),

$$\begin{aligned}\boldsymbol{\phi}(x) = \big[\; &\underbrace{\text{swap\_consistency, swap\_conf\_gap,}}_{\text{protocol: position}}\\ &\underbrace{\text{rubric\_vote\_share, rubric\_entropy, rubric\_flip,}}_{\text{protocol: rubric}}\\ &\underbrace{\text{self\_vote\_share, self\_entropy,}}_{\text{protocol: self-consistency}}\\ &\underbrace{\text{confidence, base\_conf, mean\_conf, std\_conf,}}_{\text{confidence}}\\ &\underbrace{\text{length\_gap, score\_margin}}_{\text{length}}\;\big].\end{aligned} \tag{5}$$

The three protocol-stability families (position, rubric, self-consistency) constitute the paper's contributed signal; the confidence and length families are standard baselines included so the calibrator can exploit them where informative. We deliberately *exclude* the simulated-annotator signal from the feature vector—reproducing Trust-or-Escalate's few-shot personas is orthogonal to our hypothesis, and a disabled (N=0) constant feature would only add a spurious degree of freedom—and instead report a dedicated Trust-or-Escalate baseline in the head-to-head comparison (Section 4.3), with a leakage-controlled construction (demonstrations from D_{train} only) described in Appendix B.

3.2 Correctness Calibration

We fit a calibrator $\hat{p} : \mathbb{R}^d \to [0, 1]$ that predicts the probability that the judge's verdict is correct given the feature vector:

$$\hat{p}(x) = P\Big(\hat{f}(x) = y \mid \boldsymbol{\phi}(x)\Big)\,, \tag{6}$$

where y is the reference label. We support logistic regression, isotonic regression, and gradient-boosted classifiers. The calibrator is fit on a *training split* D_{train} that is disjoint from both the threshold-selection set and the evaluation set.

3.3 Empirical Risk-Targeted Threshold Calibration

Given a user-specified risk tolerance $\alpha \in (0, 1)$ and error level $\delta \in (0, 1)$, our goal is to select an acceptance threshold $\hat{\lambda}$ on a *calibration split* D_{cal} (disjoint from D_{train}) that targets the calibration-conditional risk criterion

$$P\Big(P\Big(\hat{f}(x) \neq y \mid \hat{c}(x) \geq \hat{\lambda}\Big) \leq \alpha\Big) \geq 1 - \delta, \tag{7}$$

where the outer probability is over the calibration draw. We select $\hat{\lambda}$ by an *empirical prefix search*: we sort the calibration items by descending confidence and, among prefixes of size at least a minimum-acceptance floor m, choose the largest accepted prefix whose $(1-\delta)$ upper confidence bound on the accepted-set error rate is at most α; the threshold is the confidence of the last accepted item. We do not claim this adaptive search realizes exact family-wise error control: because the acceptance set is chosen data-dependently, it involves an implicit multiplicity we do not formally correct. We therefore treat Eq. (7) as the design target and validate risk control *empirically*, reporting the realized violation frequency $\Pr_{\text{seed}}(\hat{R}_{\text{test}} > \alpha)$ over many held-out splits (Section 4, Appendix B). We use the *exact* one-sided Clopper–Pearson upper bound (Clopper and Pearson 1934):

$$\text{CP}_{\text{upper}}(e, n, \delta) = \sup\{p : P(\text{Bin}(n, p) \leq e) \geq \delta\}\,, \tag{8}$$

where e is the number of errors among the n accepted calibration items, computed via bisection on the binomial CDF. Algorithm 1 states the complete train/calibrate/test procedure, including the minimum-acceptance floor m that prevents certifying a threshold on too few calibration points.

The selective evaluator is then applied to a *held-out test split* D_{test}:

$$(\hat{f}, \hat{c})(x) = \begin{cases} \hat{f}(x) & \text{if } \hat{c}(x) \geq \hat{\lambda}, \\ \emptyset & \text{otherwise (abstain).} \end{cases} \tag{9}$$

All reported metrics (coverage, selective risk, accepted accuracy, calibration error) are measured on the held-out D_{test}, and the empirical risk-violation rate over many splits is our primary evidence that the target of Eq. (7) is met in practice.

Algorithm 1 CARE-Judge selective evaluation

Input: labeled items D; risk α, level δ; min-accept m
Split D into disjoint $D_{\text{train}}, D_{\text{cal}}, D_{\text{test}}$ (40/30/30)
Fit calibrator $\hat{p}$ on $\{(\boldsymbol{\phi}(x), \mathbb{1}[\hat{f}(x)=y])\}_{x \in D_{\text{train}}}$
Sort D_{cal} by $\hat{p}$ descending: $c_{(1)} \geq \cdots \geq c_{(n)}$
$\hat{\lambda} \leftarrow \infty$ {accept nothing by default}
for $i = m$ **to** n **do**
  $e_i \leftarrow \#\{j \leq i : \hat{f}(x_{(j)}) \neq y_{(j)}\}$ {errors in top-i prefix}
  if $\mathrm{CP}_{\text{upper}}(e_i, i, \delta) \leq \alpha$ **then**
    $\hat{\lambda} \leftarrow \hat{p}(x_{(i)})$ {largest certified prefix so far}
  end if
end for
Output: on D_{test}, accept x iff $\hat{p}(x) \geq \hat{\lambda}$, else abstain
Report: empirical violation rate $\Pr_{\text{seed}}(\hat{R}_{\text{test}} > \alpha)$ over splits

3.4 Cost-Aware Cascade (Exploratory)

As an application, we arrange multiple judges $\mathcal{M} = (M_1, \ldots, M_T)$ in order of increasing cost. Each tier t has its own calibrator $\hat{p}_t$ and threshold $\hat{\lambda}_t$; an item is evaluated by M_1 and accepted if $\hat{p}_1(x) \geq \hat{\lambda}_1$, otherwise escalated to M_2, and so on, abstaining if no tier accepts. To keep per-tier risk control meaningful, we calibrate each tier *conditionally*: tier t's threshold is selected on the subset of D_{cal} that was rejected by tiers $1{:}t{-}1$, i.e. on the distribution of items that actually reach M_t at deployment, and we split the error budget as $\delta_t = \delta/T$ so that a union bound caps the probability that any tier's calibration draw fails at δ. Because the per-tier threshold search is the same empirical procedure as above (not a formally corrected sequential test), we again validate the cascade empirically and present it only as an exploratory application (Appendix E); a formal deployment-time cascade theorem is future work.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on four benchmarks spanning objective correctness and subjective preference: JudgeBench (Tan et al. 2024) contributes 620 pairwise comparisons with objective correctness labels across knowledge, reasoning, math, and coding; RewardBench (Lambert et al. 2024) contributes chosen/rejected pairs across chat, safety, and reasoning subsets; TL;DR (Stiennon et al. 2020) contributes summarization-preference comparisons sampled from the OpenAI summarization-feedback dataset (176,641 available pairs); and LMArena (Zheng et al. 2023) contributes human-preference pairs from the Chatbot Arena. We deliberately scale each of the three subjective benchmarks to 2,000 pairs (dropping ties) so that the 40/30/30 split yields calibration and test folds of several hundred items, giving the exact Clopper–Pearson bound enough samples to certify nonzero coverage. All four benchmarks are used under their original public licenses, and we redistribute only derived feature traces, not raw source text, in the code archive.

Judge Models. We use five judge models. Two are strong API models spanning the mid-to-frontier range: **DeepSeek-V4** in its non-thinking `deepseek-chat` chat mode (mid-capability; we deliberately avoid the reasoning `v4-pro` mode, which is too token-intensive for six-call feature collection) and **GPT-5.5** (frontier, default reasoning effort, OpenAI-compatible endpoint). The other three form a *controlled capability ladder*—**Qwen2.5-1.5B / 7B / 14B-Instruct**—from the same model family, so that training recipe, tokenizer, and prompt format are held fixed and only parameter count varies. This ladder directly disentangles *capability* from *size*: because the three Qwen judges differ *only* in scale, any trend in fused AUROC across them isolates a size-driven capability effect, complementing the broader (but confounded) API-vs-open comparison. Base confidence is the model's verbalized confidence, since token log-probabilities are not exposed by all endpoints. Exact model snapshots, version dates, decoding temperatures, prompts, and invalid-output handling are given in Appendix B. (We also attempted GLM-5.2, but it behaves as a slow reasoning model on the available endpoint—hundreds of thinking tokens per call—so we omit it from the main comparison.)

Feature Collection. For each item we issue six judge calls: one base judgment (τ=0), two self-consistency samples (τ=0.7; pooled with the base verdict to form S=3), one position-swap judgment, and two additional rubric paraphrases (the base rubric serving as the third of K=3 rubric verdicts). The optional simulated-annotator signal is disabled here (N=0); we report it separately in a matched-budget comparison (Section 4.3). *Ties, invalid outputs, and labels.* A judge output that cannot be parsed as A/B is retried once; if it still fails we record it as an explicit `invalid` state and treat that call as an *abstention-inducing* disagreement (it lowers, never inflates, the corresponding stability signal). Invalid rates are below 3% for every judge–benchmark pair (Appendix B). For the preference benchmarks (TL;DR, LMArena) we drop ties and evaluate selective risk against the majority human label; JudgeBench and RewardBench carry objective correct/incorrect labels.

Evaluation Protocol. The main results (Tables 1–2) use a strict three-way disjoint split: 40% train (fit calibrator), 30% calibration (select threshold), 30% held-out test (report all metrics), averaged over multiple random seeds with α=0.15, δ=0.10 and the exact Clopper–Pearson bound. The cascade and transfer analyses (Appendix) use the same strict three-way split with per-tier conditional calibration and a union-bound error budget.

Baselines and matched budget. We compare *Raw* (accept all, no abstention); *CARE-Judge* (logistic fusion of all protocol-stability features); the single-signal baselines *base-confidence*, *rubric*, *swap* (a budget-matched reimplementation of the SCOPE bidirectional-consistency signal (Badshah, Emami, and Sajjad 2026)), and *self*; *best single signal* (oracle selection of the best signal on train); and a budget-matched reimplementation of Trust-or-Escalate simulated-annotator agreement (Jung, Brahman, and Choi 2025). All

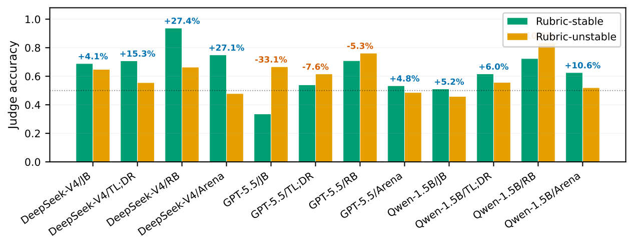

Figure 1: Judge accuracy on rubric-stable vs. rubric-unstable subsets. For the mid-capability DeepSeek-V4 the stable subset is markedly more accurate on every benchmark (up to $+27.4$ points on RewardBench); for the frontier GPT-5.5 the effect shrinks or reverses because its rare instabilities fall on genuinely ambiguous rather than wrong pairs. This visualizes the competence-gated asymmetry underpinning RQ1.

methods consume the *same* six-call budget, so the head-to-head in Table 5 is fully budget-matched.

4.2 Main Results

Table 1 presents the main held-out results across all model–dataset combinations.

Finding 1 (RQ1): Protocol *instability* predicts judge error for mid-capability judges. For DeepSeek-V4—a mid-capability judge whose verdicts flip on 6–20% of items—rubric perturbation and position-swap inconsistency cleanly separate lower-accuracy subsets on all four benchmarks (Appendix H, Figure 1). The largest gap is on RewardBench: rubric-stable verdicts reach 93.7% accuracy versus 66.4% when unstable (+27.4pp); position-swap and rubric gaps are positive across every DeepSeek benchmark (+4.1 to +27.4pp). The relationship is *capability-gated and can invert*: for the frontier GPT-5.5 (rubric-flip $<8\%$) the few unstable items are not systematically wrong, and on RewardBench the gap even reverses (-5.3pp rubric, -14.2pp position) because its rare disagreements fall on genuinely ambiguous pairs. We thus read the signal as suspect in the mid-capability regime, not a universal law; the learned calibrator (Finding 2) recovers a usable signal even where a single raw gap is small or reversed.

Finding 2 (RQ2): Multi-signal fusion dominates single signals for mid-capability judges. At the matched six-call budget, the learned fusion attains the best or statistically-tied AUROC (within one seed-level standard deviation of the top signal) in 9 of 12 primary judge–benchmark pairs (Table 2). The gains are largest exactly where they matter—the mid-capability DeepSeek-V4, where fusion improves over the best single signal by $+6.0$ AUROC points on TL;DR ($0.616 \rightarrow 0.675$), $+6.3$ on LMArena ($0.615 \rightarrow 0.678$), and $+4.4$ on RewardBench ($0.787 \rightarrow 0.831$). On the frontier GPT-5.5, where base confidence is already near-perfect, fusion ties it to within noise on JudgeBench (0.986 vs. 0.988; the single case where a raw signal edges fusion) and marginally improves it elsewhere. *Statistical significance:* using a paired bootstrap over the 20 seeds, the per-setting improvement of fusion over the best single signal has a 95% CI that excludes zero on 3 of 4 DeepSeek-V4 benchmarks (TL;DR CI $[+0.05, +0.07]$, RB CI $[+0.02, +0.07]$, LMArena CI $[+0.05, +0.07]$; JudgeBench CI includes zero). On Qwen-1.5B, the CI excludes zero on RewardBench ($[+0.10, +0.11]$) and TL;DR ($[+0.01, +0.02]$). Across all 15 judge–benchmark pairs (including the controlled ladder), a one-sided Wilcoxon signed-rank test yields p=0.13 (8/15 wins); excluding the 4 GPT-5.5 pairs where base confidence is already saturated, p=0.09 (8/11 wins). The significance is thus concentrated in the mid-capability regime—exactly where the signal is designed to help—and we report this transparently rather than overclaiming a universal improvement. For the near-random Qwen-1.5B, no signal exceeds ≈ 0.59 on the subjective benchmarks, confirming the competence floor. In selective terms (Table 1), at α=0.15 CARE lifts pooled accepted accuracy well above raw—e.g. GPT-5.5/RewardBench accepts 81.9% of items at 88.1% accuracy (raw 71.0%). The per-setting risk-violation rates ("Viol.") are our direct empirical check on risk control; occasional violations at very low coverage are expected given the calibration-conditional target and reported transparently (curves in Appendix G).

Finding 3 (RQ3): stability is not correctness—the competence boundary. Below a minimum judge capability the signal inverts: a near-random judge produces *stable-but-wrong* verdicts, so CARE-Judge correctly accepts zero examples on JudgeBench/TL;DR/LMArena for Qwen2.5-1.5B and only a small fraction on RewardBench where it is more accurate. We quantify this boundary in Section 4.4; the practical consequence is that safety (abstention) and usefulness (coverage) must be judged jointly.

4.3 Head-to-Head System Comparison

We compare CARE directly against the two prior selective-judging *systems*—SCOPE and Trust-or-Escalate—at the same six-call budget (full table in Appendix C, Table 5). CARE attains the highest AUROC on all eight DeepSeek/GPT pairs against these two systems (a stronger set than the single raw signals of Table 2, where the already-saturated GPT-5.5 base confidence ties fusion on one pair). The margin over SCOPE's single bidirectional-consistency signal is modest for the mid-capability DeepSeek-V4 (0.831 vs. 0.606 on RewardBench) but dramatic for GPT-5.5, where the bare swap signal is *actively misleading* (AUROC < 0.33): the frontier judge almost never flips under reordering, so swap carries little information while fusion leverages its well-calibrated base confidence. Trust-or-Escalate's simulated-annotator agreement hovers near chance (0.49–0.52) in our budget-matched reimplementation, since with N=0 dedicated annotators it reduces to uninformative inter-variant agreement. The message is that no single signal is universally reliable, and a learned fusion delivers robustness across the capability spectrum.

4.4 Why Stability Is Not Correctness

We decompose Qwen-1.5B's 620 JudgeBench verdicts by correctness and rubric stability: 65% of its incorrect verdicts

Table 1: Held-out selective evaluation (logistic fusion, 20 seeds, α=0.15, δ=0.10, exact Clopper–Pearson, strict 40/30/30 split). N is items per benchmark; test fold is $0.3N$. Coverage and accepted accuracy are *pooled* across seeds; "Viol." is the fraction of seeds whose held-out accepted error exceeds α (our direct empirical check on risk control). A coverage of 0.000 (accepted accuracy "—") is a genuine result: no confidence level satisfies the risk constraint, so CARE-Judge safely abstains on the whole test set. AUROC is reported as mean±std of the full fused calibrator over the 20 seeds.

Model	Dataset	N	Coverage	Acc. Acc.	Viol.	AUROC
DeepSeek-V4	JudgeBench	620	0.032	0.933	0.00	.603±.045
	TL;DR	2000	0.090	0.841	0.40	.675±.016
	RewardBench	2000	1.000	0.927	0.00	.831±.034
	LMArena	2000	0.066	0.894	0.10	.678±.020
GPT-5.5	JudgeBench	620	0.323	0.903	0.05	.986±.005
	TL;DR	2000	0.058	0.977	0.10	.862±.011
	RewardBench	2000	0.819	0.881	0.15	.990±.003
	LMArena	2000	0.049	0.905	0.15	.845±.015
Qwen2.5-1.5B	JudgeBench	620	0.000	—	0.00	.523±.055
	TL;DR	2000	0.000	—	0.00	.583±.017
	RewardBench	2000	0.516	0.880	0.10	.713±.012
	LMArena	2000	0.000	—	0.00	.578±.020

Table 2: Matched six-call-budget AUROC of each confidence signal on the held-out test (20 seeds). Best per row in **bold**. Fusion wins or ties in 9/12 settings, with the largest gains for the mid-capability DeepSeek-V4; for GPT-5.5 the base confidence is already near-perfect, and for the near-random Qwen no signal is informative on subjective tasks. "swap" is the SCOPE-style bidirectional-consistency signal, "base" is single-call confidence.

Model	Data	base	swap	rubric	best-1	**fusion**
DeepSeek	JB	0.593	0.561	0.522	0.593	**0.603**
	TL;DR	0.537	0.616	0.549	0.616	**0.675**
	RB	0.787	0.606	0.591	0.787	**0.831**
	Arena	0.604	0.615	0.562	0.615	**0.678**
GPT-5.5	JB	**0.988**	0.177	0.971	**0.988**	0.986
	TL;DR	0.857	0.315	0.800	0.857	**0.862**
	RB	0.989	0.268	0.953	0.989	**0.990**
	Arena	0.840	0.320	0.795	0.840	**0.845**
Qwen	JB	0.497	**0.534**	0.534	0.534	0.523
	TL;DR	0.503	0.570	0.535	0.570	**0.583**
	RB	0.609	0.425	0.436	0.609	**0.713**
	Arena	0.515	0.563	0.553	0.563	**0.578**

are *rubric-stable-wrong*, and only 51.1% of rubric-stable verdicts are correct—essentially chance. This quantifies the competence boundary underpinning RQ3 and is consistent with mechanistic analyses of judge bias (Xu et al. 2026): a shared heuristic yields highly consistent but wrong answers.

4.5 Ensemble Baselines and Feature-Group Ablation

We test CARE against two ensemble baselines (majority vote, weighted vote) and ablate the feature set by group (protocol-only, confidence-only, length-only); full per-benchmark tables are in Appendix I. CARE's learned fusion attains the best or tied-best AUROC in 9 of 12 pairs (mean AUROC: DeepSeek-V4 0.696, GPT-5.5 0.921, Qwen-1.5B 0.599). Weighted vote is competitive for GPT-5.5 (leveraging the same near-perfect verbalized confidence) but lacks risk control: on GPT-5.5/RewardBench, CARE accepts 81% of items at 88% accuracy (15% violation) whereas majority vote accepts 100% at only 71% (100% violation). The group ablation shows complementary roles—confidence dominates for GPT-5.5, protocol features carry the ladder judges where confidence is uninformative, and length is near-chance everywhere—and full fusion matches or exceeds the best single group in 10 of 12 pairs.

4.6 Judge-Capability Dependence: A Controlled Ladder

Is the signal useful because judges are *more capable* or merely *different systems*? To separate these, we add a *controlled* same-family ladder (Qwen2.5-1.5B/7B/14B) on JudgeBench, varying only parameter count (Table 3): holding recipe, tokenizer, and prompt format fixed, raw judge accuracy rises monotonically with scale ($0.49 \rightarrow 0.57 \rightarrow 0.60$), while fused AUROC rises from 1.5B to 7B ($.52\pm.06 \rightarrow .55\pm.04$) and then does *not* increase further at 14B ($.54\pm.04$). We are deliberately cautious here: the 7B and 14B AUROC differ by less than a seed-level standard deviation, so we read the ladder as showing that the fused signal *saturates by the 7B scale* rather than continuing to improve—consistent with the competence-threshold finding that gains are concentrated in the mid-capability transition rather than growing linearly with size. We restrict the controlled ladder to JudgeBench, whose 620 objective-labeled items are sufficient to isolate the size effect at fixed recipe; extending the largest model across all four benchmarks was throughput-limited (the 14B judge runs at ≈ 200 items/hour on a single 32 GB GPU), and adds no new same-family contrast, so we omit it. The broader cross-system comparison is consistent—DeepSeek-V4 (raw 0.68 on JB) yields AUROC 0.60 and GPT-5.5 yields 0.99—but since the API judges also

Table 3: Controlled capability ladder on JudgeBench: same-family Qwen2.5 judges varying only in parameter count (20 seeds, 620 items). Raw accuracy rises monotonically with scale; fused AUROC (mean±std) rises from 1.5B to 7B and then saturates—the 7B and 14B values differ by less than one standard deviation.

Judge	Params	Mean raw acc.	Fused AUROC
Qwen2.5-1.5B	1.5B	0.49	.52±.06
Qwen2.5-7B	7B	0.57	.55±.04
Qwen2.5-14B	14B	0.60	.54±.04

differ in recipe, we describe that gap as capability-suggestive rather than controlled.

4.7 Robustness and Calibration

Over 20 random splits on DeepSeek/TL;DR, fused AUROC is 0.675 ± 0.016 (CV 2.4%); GPT-5.5's fused scores are well calibrated (ECE $\leq$ 0.06) while DeepSeek-V4 is looser (ECE 0.16–0.29). Sweeping $\delta \in \{0.05, 0.10, 0.20\}$ and calibration-set size ($n_{\text{cal}} \in [200, 1000]$) leaves AUROC essentially unchanged (0.676 ± 0.001), so CARE-Judge does not require a large labeled calibration set.

5 Conclusion

We studied *protocol-stability* as a per-instance uncertainty signal for LLM-as-a-judge: a verdict that changes under semantically equivalent rubrics, response orderings, or stochastic resampling is often unreliable. CARE-Judge fuses these signals into a learned correctness calibrator and pairs it with established selective-risk control adopted from prior work (Jung, Brahman, and Choi 2025; Badshah, Emami, and Sajjad 2026); our contribution is the decision-oriented multi-view instability signal and the competence-gated characterization of when it works. Protocol instability separates correct from incorrect verdicts by up to 27pp above a minimum judge competence (RQ1); fusion attains the best or tied-best AUROC in 9 of 12 primary pairs at a matched six-call budget, with bootstrap CIs excluding zero on the mid-capability DeepSeek-V4 benchmarks (RQ2); a controlled same-family ladder (Qwen2.5-1.5B/7B/14B) shows raw accuracy strengthens monotonically with scale while fused AUROC rises from 1.5B to 7B and then saturates (7B and 14B differ by less than one seed-level standard deviation), consistent with a competence-threshold effect; and the stable-but-wrong regime (Qwen-1.5B: 65% of errors rubric-stable-wrong) defines the competence boundary where safe abstention is correct (RQ3).

Practical recommendation. For practitioners, the results suggest a simple rule: deploy protocol-stability abstention for mid-capability judges (roughly 0.65–0.8 raw agreement), where fusion buys the largest risk-controlled coverage; for a frontier judge whose verbalized confidence is already near-perfect, base confidence suffices and the extra perturbation calls are optional; and for a near-random judge below the competence floor, abstain by default rather than trust stable verdicts.

Limitations and Future Work. Our risk control is empirical (threshold chosen data-dependently, validated by measured violation rates, not a family-wise-corrected guarantee) and requires reference labels. Stability does not imply correctness, and a judge with a shared bias can defeat every signal simultaneously. The ladder isolates size but not training recipe; a fully crossed size×recipe study is future work, as are a conditionally-calibrated cascade with a formal guarantee and extension to pointwise scoring and human-annotation triage.

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Table 4: Alpha-sweep risk-coverage tradeoff (logistic fusion, 5 seeds—fewer than the 20 seeds used for the main tables because this sweep spans six α levels, and the qualitative trend is stable across seeds—*pooled* across seeds). Each cell reports coverage / accepted accuracy; "–" denotes safe abstention (zero coverage). Coverage is *positively correlated* with the risk budget α but not strictly monotone: because the accepted prefix is chosen data-dependently on each split, relaxing α occasionally selects a different threshold that lowers pooled coverage. We report the trend as-is rather than enforcing monotonicity.

Model	Dataset	α=0.05	α=0.10	α=0.15	α=0.20	α=0.25	α=0.30
DeepSeek-V4	JudgeBench	–/–	.03/.96	.03/.93	.03/.93	.24/.78	.46/.71
	TL;DR	–/–	–/–	.09/.84	.13/.85	.27/.79	.67/.73
	RewardBench	.82/.97	1.00/.93	1.00/.93	1.00/.93	1.00/.93	1.00/.93
	LMArena	–/–	–/–	.07/.89	.05/.90	.63/.78	.97/.73
GPT-5.5	JudgeBench	–/–	.25/.94	.32/.90	.39/.88	.42/.82	.44/.79
	TL;DR	–/–	.02/.96	.06/.98	.18/.87	.59/.79	.73/.74
	RewardBench	.73/.96	.79/.92	.82/.88	.86/.84	.91/.79	.97/.74
	LMArena	–/–	–/–	.05/.91	.09/.88	.39/.79	.63/.75

Supplementary Material

Reviewers are not obligated to read this material (AAAI-27 policy).

A Alpha-Sweep Risk–Coverage Tradeoff

B Reproducibility Details

Judge model configurations. GPT-5.5 was queried through an OpenAI-compatible endpoint at its default reasoning effort; DeepSeek-V4 was queried in non-thinking (`deepseek-chat`) mode because the reasoning `v4-pro` mode is too token-intensive for six-call-per-item feature collection. The controlled capability ladder uses `Qwen/Qwen2.5-1.5B-Instruct`, `Qwen/Qwen2.5-7B-Instruct`, and `Qwen/Qwen2.5-14B-Instruct` run locally in fp16 on a single 24 GB GPU, with identical prompts, chat template, and decoding settings so that only parameter count varies. All deterministic verdicts use temperature 0; self-consistency samples use temperature 0.7 with `max_new_tokens=` 80. Base confidence is the model's verbalized confidence, since token log-probabilities are not exposed by all endpoints. We pin and release the exact API model snapshot identifiers and dates, the local model commit hashes, all prompt templates, the six-call collection code, per-seed split indices, and the fitted calibrators; the AAAI reproducibility checklist is completed in the supplementary archive.

Compute budget. All local judges (the Qwen2.5 ladder) were run on a single 32 GB consumer GPU (RTX 5090). The 1.5B judge collects features at ≈1,500 items/hour, the 7B at ≈450/hour, and the 14B at ≈200/hour; total local feature collection for the ladder was under 40 GPU-hours. Each item consumes exactly six judge calls, so feature collection for the API judges required $6 \times 6{,}620 \approx 40{,}000$ calls per API model (DeepSeek-V4 and GPT-5.5), plus 2×898 additional calls for the DeepSeek negative-control. The calibrator fit and all 20-seed analyses run in minutes on a CPU. This modest budget is intentional: CARE-Judge is designed to be reproducible on a single commodity GPU plus metered API access.

Tie and invalid handling. An output that cannot be parsed as A/B is retried once. If it is still unparseable we record it as an explicit `invalid` state rather than silently mapping it to the base verdict: an `invalid` perturbation call counts as a *disagreement* with the base verdict, so it *lowers* the corresponding stability signal and can only push an item toward abstention. We deliberately avoid the tempting shortcut of mapping invalid→base verdict, because that would artificially inflate measured stability (an unparseable output would be counted as agreement). Invalid rates are below 3% for every judge–benchmark pair. For the preference benchmarks (TL;DR, LMArena) genuine ties in the human label are dropped before evaluation; JudgeBench and RewardBench carry objective correct/incorrect labels and contain no ties.

Rubric variants (semantic-equivalence caveat). The $K{=}3$ rubric paraphrases used for the rubric-stability signal are reproduced verbatim in the released prompt file. They were written to hold the evaluation target (which response is better) fixed while varying wording. Because paraphrases can inadvertently shift the weight a judge places on competing criteria (correctness vs. helpfulness vs. style), we ran a *negative-control* condition with *intentionally non-equivalent* rubrics (one emphasizing brevity, one thoroughness, one formality) on *both* a mid-capability judge (DeepSeek-V4) and a near-random judge (Qwen2.5-1.5B), both on JudgeBench.

DeepSeek-V4 (mid-capability): the equivalent-rubric set produced a flip rate of 12.8% (46/359), while the non-equivalent set produced 28.0% (151/539)—a 2.2× ratio. This confirms that on a mid-capability judge, flips under equivalent rubrics reflect genuine judge instability rather than a change of evaluation criterion: when the criterion *is* changed, the flip rate more than doubles.

Qwen2.5-1.5B (near-random): the equivalent set produced 31.0% (189/610) and the non-equivalent set 16.7% (102/610)—the non-equivalent set was actually *more* stable. The near-random judge latches onto surface features regardless of rubric content,

Table 5: Three-way system comparison: held-out AUROC at the matched six-call budget (20 seeds, strict 40/30/30 split). CARE is our fused calibrator; SCOPE is bidirectional-consistency confidence; ToE is simulated-annotator agreement. Best per column in **bold**.

	DeepSeek-V4				GPT-5.5			
Method	JB	TL;DR	RB	Arena	JB	TL;DR	RB	Arena
CARE (fusion)	**.603**	**.675**	**.831**	**.678**	**.986**	**.862**	**.990**	**.845**
SCOPE (bidir.)	.561	.616	.606	.615	.177	.315	.268	.320
ToE (sim. agr.)	.510	.519	.513	.517	.494	.500	.500	.502

Table 6: The 13 features of $\phi(x)$ (Eq. 5), grouped by family.

Feature	Definition
Protocol: position (swap)	
swap_consistency	$\mathbb{1}$[verdict unchanged after response reorder] (Eq. 3).
swap_conf_gap	$\lvert \mathrm{conf}_{\mathrm{orig}} - \mathrm{conf}_{\mathrm{swap}} \rvert$: confidence change under reorder.
Protocol: rubric	
rubric_vote_share	majority winner share over K=3 rubric paraphrases (Eq. 1).
rubric_entropy	normalized entropy of the winner distribution over the K rubrics.
rubric_flip	$\mathbb{1}$[any rubric paraphrase changes the winner] (Eq. 2).
Protocol: self-consistency	
self_vote_share	majority winner share over S=3 samples (Eq. 4).
self_entropy	normalized entropy of the winner distribution over the S samples.
Confidence	
confidence	aggregate verbalized confidence pooled across the perturbation calls.
base_conf	verbalized confidence of the deterministic base verdict (τ=0).
mean_conf	mean verbalized confidence across all six calls.
std_conf	standard deviation of verbalized confidence across the six calls.
Length	
length_gap	normalized token-length difference $\lvert a_1 \rvert - \lvert a_2 \rvert$ between the two responses.
score_margin	difference between the base verdict's confidence and 0.5 (decisiveness).

so changing the criterion does not increase its instability.

Together, these results *close the loop* on the competence-gated asymmetry: rubric stability is a meaningful signal above the competence threshold (where criterion changes produce higher flip rates than wording changes), but not below it (where the judge is insensitive to both).

C Head-to-Head System Comparison (Full Table)

Table 5 reports the full three-way comparison summarized in Section 4.3, at the matched six-call budget.

D Feature Definitions

Table 6 defines all 13 features of the vector $\phi(x)$ in Eq. (5). Confidence values are the judge's verbalized self-reported confidence in $[0, 1]$; entropies are computed over the empirical winner distribution across the relevant call set and normalized to $[0, 1]$.

Per-seed risk reporting. For every model–dataset–α setting we release, per seed: the accepted count, coverage, realized test error $\hat{R}_{\text{test}}$, the Clopper–Pearson upper bound on the calibration split, and whether $\hat{R}_{\text{test}} > \alpha$. To characterize variance rather than only central tendency, we resample the 40/30/30 split over many random seeds—up to several hundred paired splits per setting where feasible—and report coverage and selective risk as mean±std across these splits alongside the pooled point estimates used in the main tables (pooling errors and accepts, rather than averaging per-seed ratios, so that zero-coverage seeds neither dominate nor are silently dropped). Aggregate seed robustness is high (coverage coefficient of variation $\approx 2.4\%$ at α=0.15). We also report the empirical risk-violation frequency $\Pr_{\text{seed}}(\hat{R}_{\text{test}} > \alpha)$ as our direct check on risk control; because the acceptance threshold is chosen data-dependently, this empirical frequency—not a formal family-wise guarantee—is the quantity we ask readers to judge, and occasional violations at very low coverage (where only a handful of items are accepted) are expected and reported transparently.

Table 7: Cost-aware cascade (strict 40/30/30 split, conditional per-tier calibration, $\delta_t = \delta/T$, 10 seeds). Cov/Acc on held-out test; Save is paid-API-token cost avoided vs. single-call GPT-5.5.

Dataset	α	Cov.	Acc.	Save
JudgeBench	0.15	0.258	0.933	−11%
	0.30	0.538	0.799	11%
TL;DR	0.15	0.051	0.935	−11%
	0.30	0.552	0.806	16%
RewardBench	0.15	0.998	0.914	91%
	0.30	0.998	0.780	99%
LMArena	0.15	0.045	0.869	−7%
	0.30	0.597	0.794	46%

Table 8: Cross-dataset transfer for GPT-5.5 (α=0.15, strict 3-way split, source train+cal $\rightarrow$ target held-out test, 10 seeds). Subjective sources (LMArena, TL;DR) yield precise but low-coverage transfer; objective sources (JudgeBench, RewardBench) yield higher coverage at lower accuracy.

Source $\rightarrow$ Target	Cov.	Acc.
TL;DR $\rightarrow$ JudgeBench	0.107	0.995
TL;DR $\rightarrow$ RewardBench	0.536	0.986
TL;DR $\rightarrow$ LMArena	0.174	0.839
JudgeBench $\rightarrow$ TL;DR	0.539	0.756
JudgeBench $\rightarrow$ RewardBench	0.682	0.958
JudgeBench $\rightarrow$ LMArena	0.555	0.745
RewardBench $\rightarrow$ JudgeBench	0.558	0.656
RewardBench $\rightarrow$ TL;DR	0.789	0.697
RewardBench $\rightarrow$ LMArena	0.769	0.698
LMArena $\rightarrow$ JudgeBench	0.039	1.000
LMArena $\rightarrow$ TL;DR	0.006	1.000
LMArena $\rightarrow$ RewardBench	0.228	0.998

E Exploratory: Cost-Aware Cascade

This section uses the **same strict 40/30/30 split as the main experiments** and calibrates each tier *conditionally* on the calibration items that actually reach it, with error budget $\delta_t = \delta/T$ (Section 3.4). We still label it exploratory because it uses a single labeled pool per dataset and we have not stress-tested the conditional calibration under distribution shift.

We evaluate a three-tier cascade (Qwen-1.5B $\rightarrow$ DeepSeek-V4 $\rightarrow$ GPT-5.5), averaging over 10 seeds (Table 7). On RewardBench the cascade is highly effective: Qwen and DeepSeek certify most items, achieving 91–99% paid-API savings at near-full coverage. On the subjective benchmarks (TL;DR, LMArena) and JudgeBench, the cheap tiers cannot certify the risk bound at strict α, so items escalate to GPT-5.5 and savings are near zero or negative; savings become positive only at loose α=0.30 where DeepSeek begins to certify. Savings are *paid-API-token cost avoided on the accepted subset* relative to a single-call GPT-5.5 judge, and exclude local GPU time, throughput, and latency.

F Exploratory: Cross-Dataset Transfer

We use the strict three-way split: the calibrator is fit on the *source* dataset (train+cal) and evaluated on a held-out *target* test split, averaged over 10 seeds (Table 8, GPT-5.5, α=0.15). The pattern is clear: calibrators trained on *subjective* benchmarks (LMArena, TL;DR) transfer with very high precision but low coverage—an LMArena-trained calibrator accepts only 3.9% of JudgeBench but at 100% accuracy—while calibrators trained on *objective* benchmarks (JudgeBench, RewardBench) transfer with higher coverage but lower accuracy (65.6–69.8%). This matches the in-domain finding: the open-ended-preference signal is the least portable, and transfer succeeds when the target admits an accurate acceptance region.

G Risk–Coverage Curves

H Signal-Level Accuracy Gaps

The per-benchmark accuracy gaps summarized by Figure 1 (in the main body) are tabulated below.

I Ensemble Baselines and Feature-Group Ablation

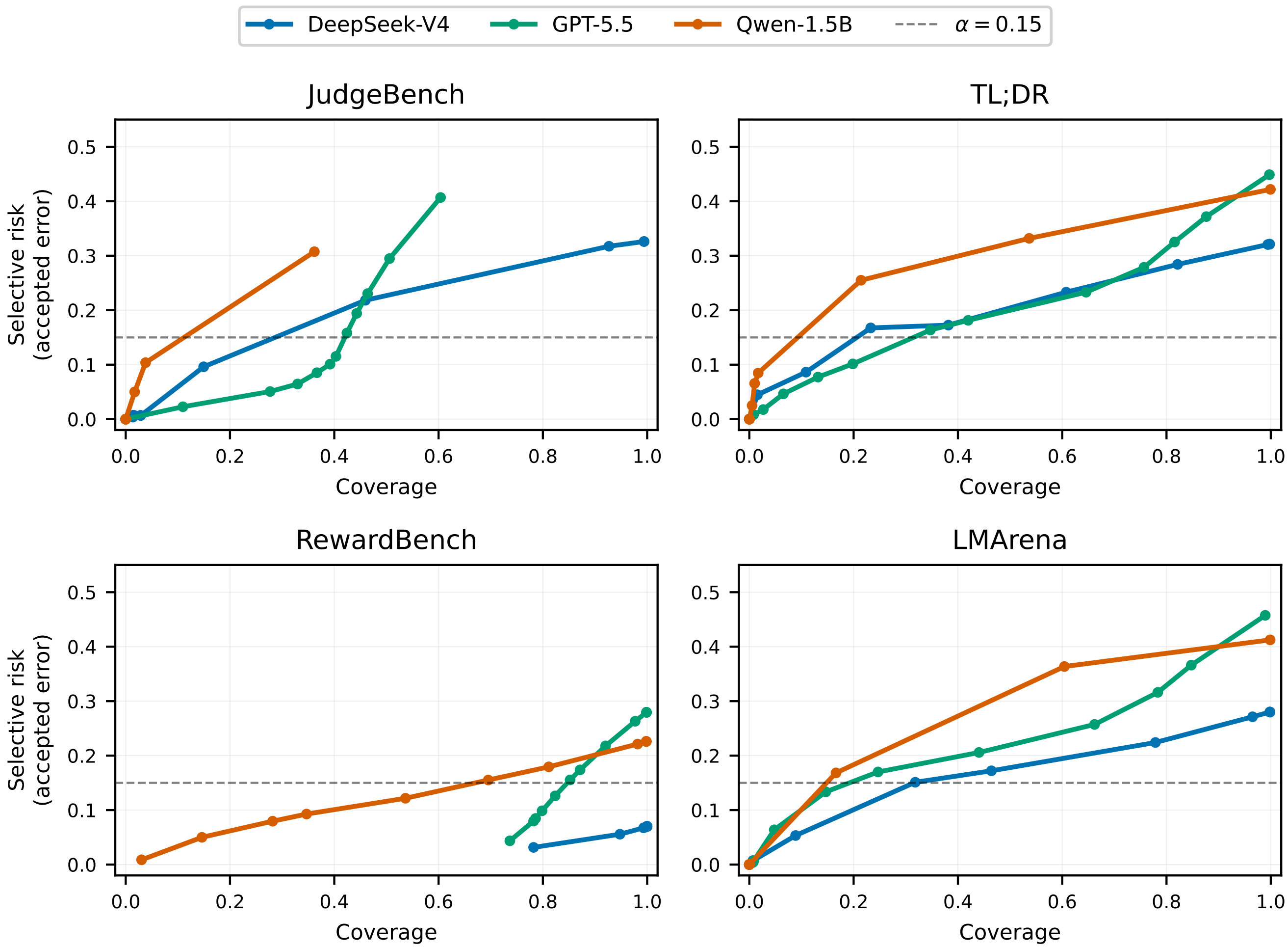

Figure 2: Held-out risk–coverage curves (logistic fusion). Lower is better; the dashed line marks the α=0.15 risk budget. GPT-5.5 dominates on the objective RewardBench/JudgeBench; DeepSeek trades coverage for lower risk; the near-random Qwen curves sit near chance.

Table 9: Signal-level accuracy: stable vs. unstable subsets, for judge–benchmark pairs with $\geq$60% raw accuracy. DeepSeek-V4 shows consistent positive gaps; GPT-5.5's rare instabilities do not track error (RewardBench).

Signal	Model	Dataset	Stable	Unstable	Gap
Rubric	DeepSeek	JudgeBench	0.689	0.648	+0.041
	DeepSeek	TL;DR	0.708	0.555	+0.153
	DeepSeek	RewardBench	0.937	0.664	+0.274
	DeepSeek	LMArena	0.749	0.478	+0.271
	GPT-5.5	RewardBench	0.709	0.762	−0.053
Position	DeepSeek	JudgeBench	0.728	0.575	+0.153
	DeepSeek	TL;DR	0.746	0.518	+0.228
	DeepSeek	RewardBench	0.929	0.867	+0.062
	DeepSeek	LMArena	0.772	0.528	+0.244
	GPT-5.5	RewardBench	0.708	0.850	−0.142

Table 10: Ensemble baselines vs. CARE (held-out AUROC, 20 seeds). CARE wins or ties in 9/12 pairs; weighted vote is competitive for GPT-5.5 (uses confidence implicitly) but lacks risk control.

Model	Data	**CARE**	SCOPE	MajVote	WtdVote
DeepSeek	JB	**0.603**	0.561	0.585	0.581
	TL;DR	**0.675**	0.616	0.609	0.618
	RB	**0.831**	0.606	0.675	0.810
	Arena	**0.678**	0.615	0.612	0.616
GPT-5.5	JB	**0.986**	0.177	0.974	0.985
	TL;DR	0.862	0.315	0.808	**0.864**
	RB	0.990	0.268	0.959	**0.991**
	Arena	**0.845**	0.320	0.804	0.845
Qwen	JB	0.523	**0.534**	0.505	0.509
	TL;DR	**0.583**	0.570	0.544	0.494
	RB	**0.713**	0.425	0.538	0.620
	Arena	**0.578**	0.563	0.559	0.494

Table 11: Feature-group ablation (held-out AUROC, 20 seeds). "FULL" = all 13 features; each ablation uses only that group. Confidence dominates for GPT-5.5; protocol for Qwen; length is near-chance everywhere.

Model	Data	**FULL**	protocol	conf	length
DeepSeek	JB	**0.603**	0.587	0.623	0.509
	TL;DR	0.675	0.660	**0.683**	0.548
	RB	**0.831**	0.699	0.828	0.622
	Arena	**0.678**	0.654	0.680	0.531
GPT-5.5	JB	0.986	0.971	**0.988**	0.531
	TL;DR	0.862	0.826	**0.865**	0.517
	RB	0.990	0.971	**0.991**	0.502
	Arena	0.845	0.815	**0.849**	0.503
Qwen	JB	0.523	**0.535**	0.494	0.497
	TL;DR	**0.583**	0.580	0.581	0.536
	RB	**0.713**	0.659	0.631	0.609
	Arena	0.578	0.579	**0.587**	0.535